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VISIONMATE AI

An Intelligent Voice-based Assistant For
Visually Impaired People

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Project Guide :

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Batch : 05

Phase I | Review II | 06.10.2026

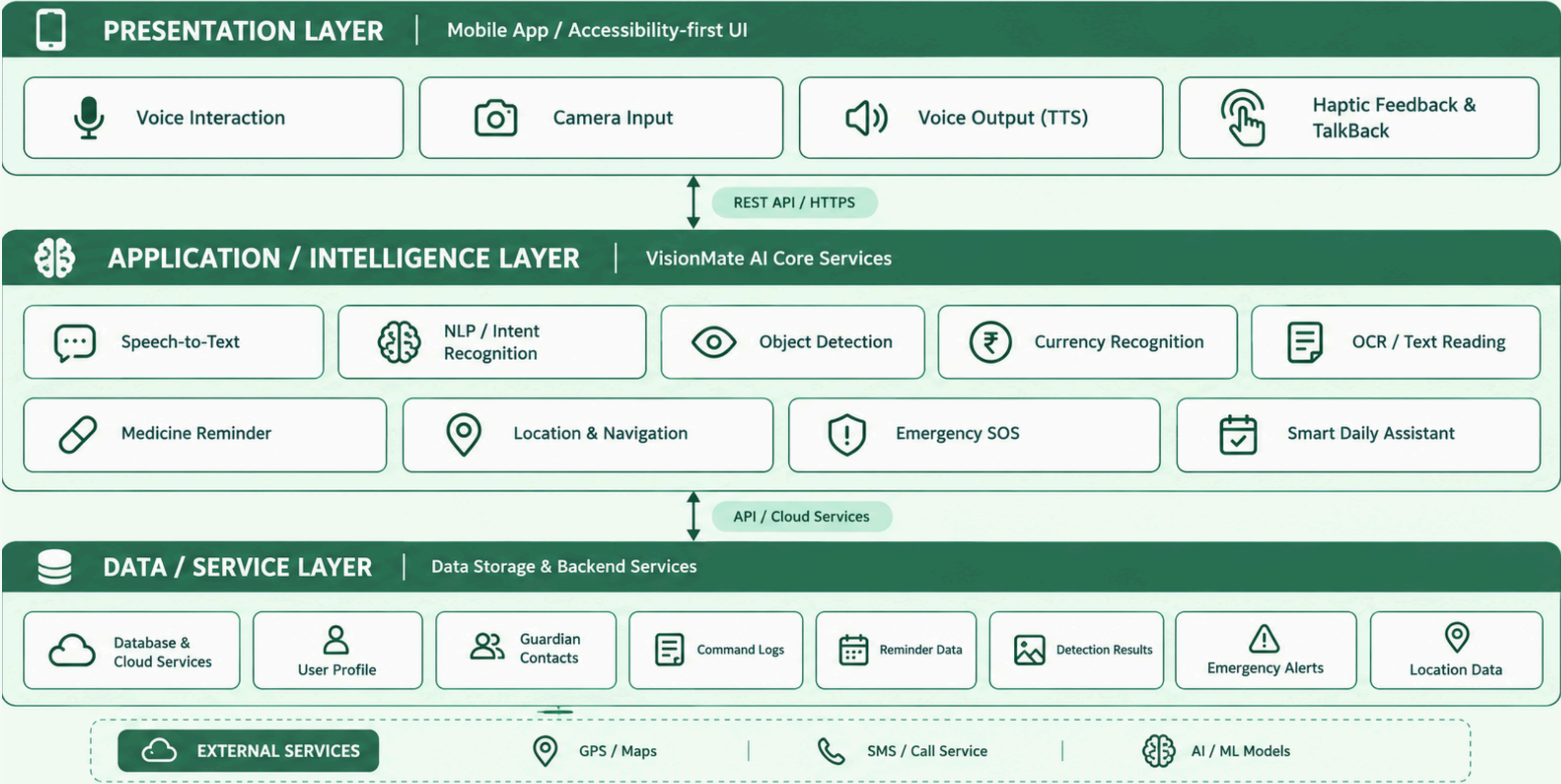


Project Overview

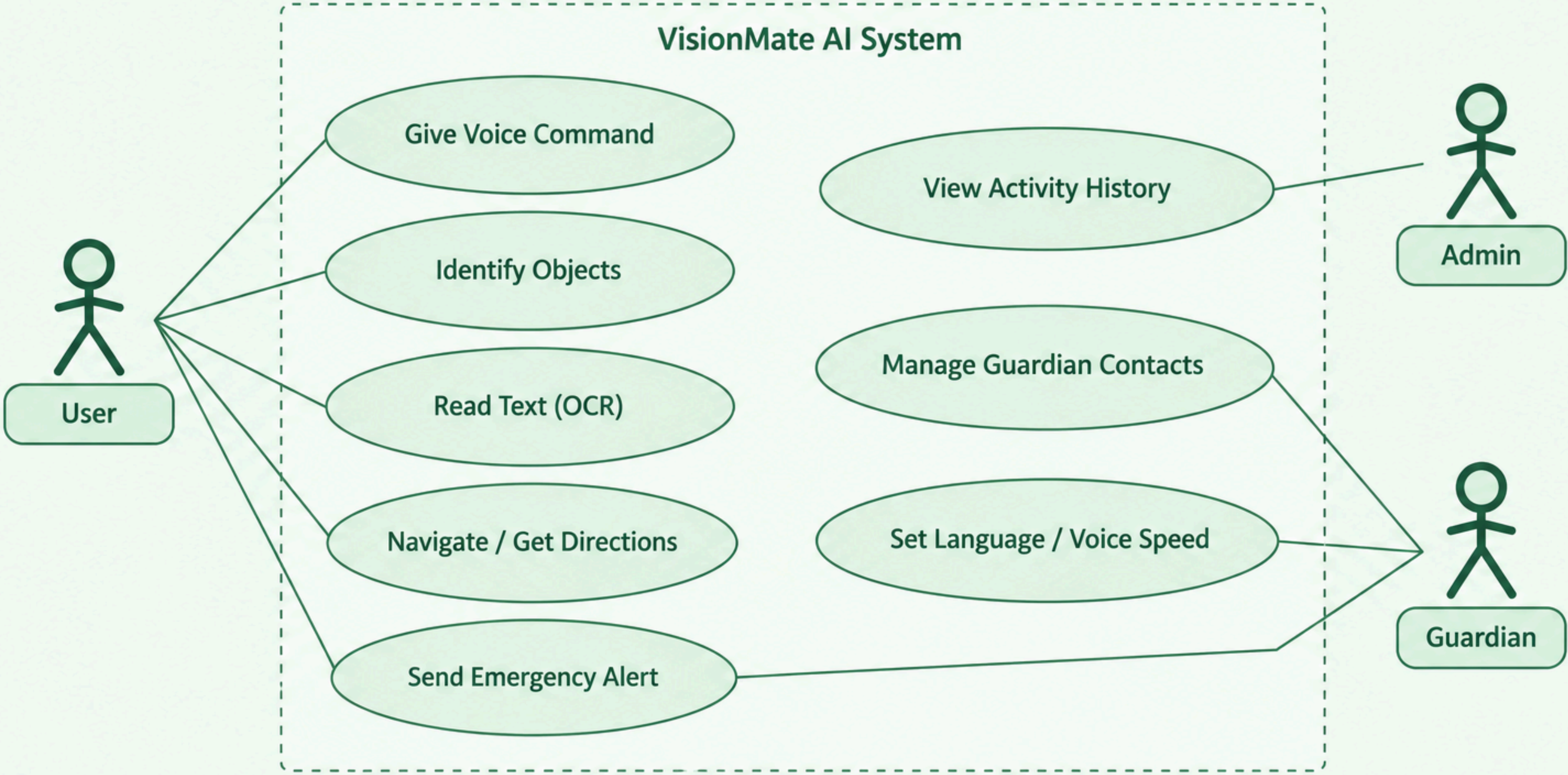


- **Problem:** Visually impaired people face difficulties in identifying objects, reading text, recognizing currency, navigating locations, and getting immediate help during emergencies. Existing solutions may require multiple apps or assistance from others.
- **Solution:** VisionMate AI is an intelligent voice-based assistant that combines multiple accessibility features in a single system.
- **Objectives:**
 1. To provide voice-based interaction for easy and hands-free access.
 2. To detect and identify objects and obstacles around the user.
 3. To recognize currency notes accurately.
 4. To read and convert printed text into speech.

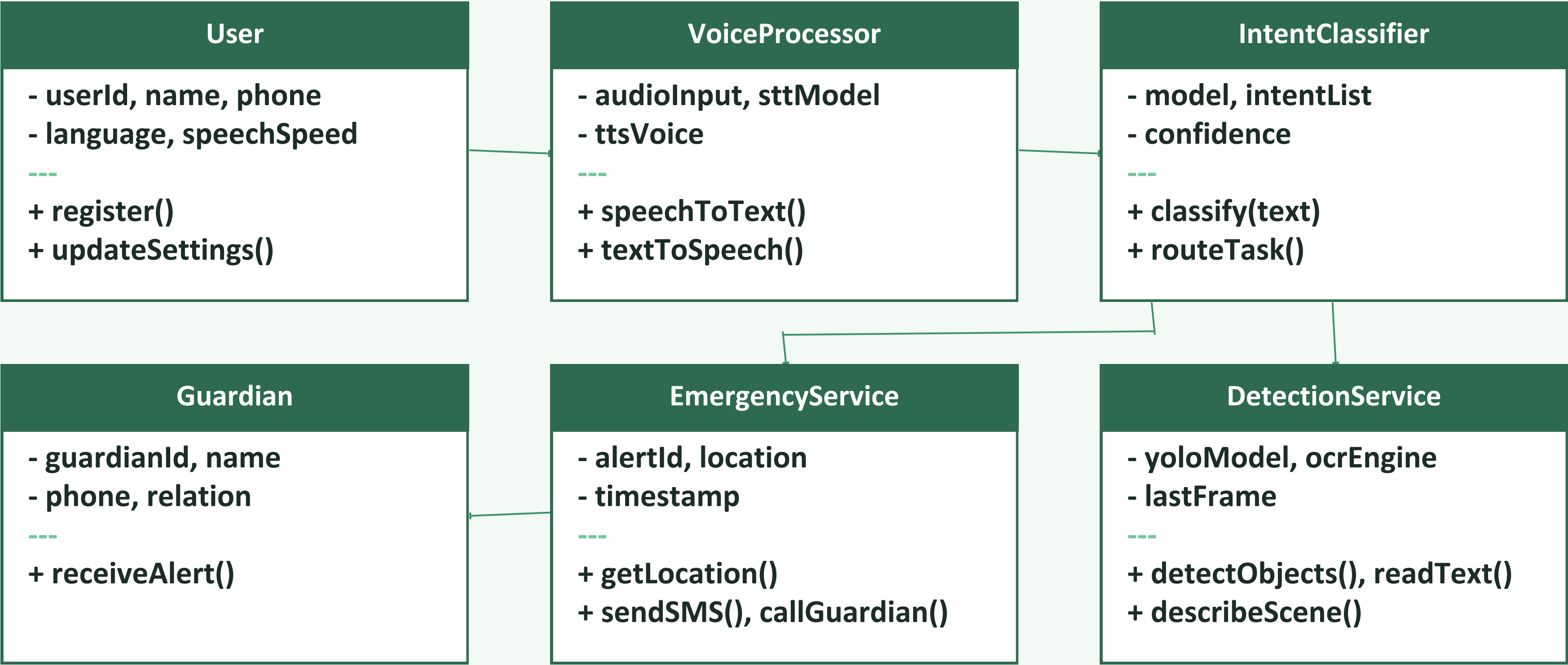
System Architecture



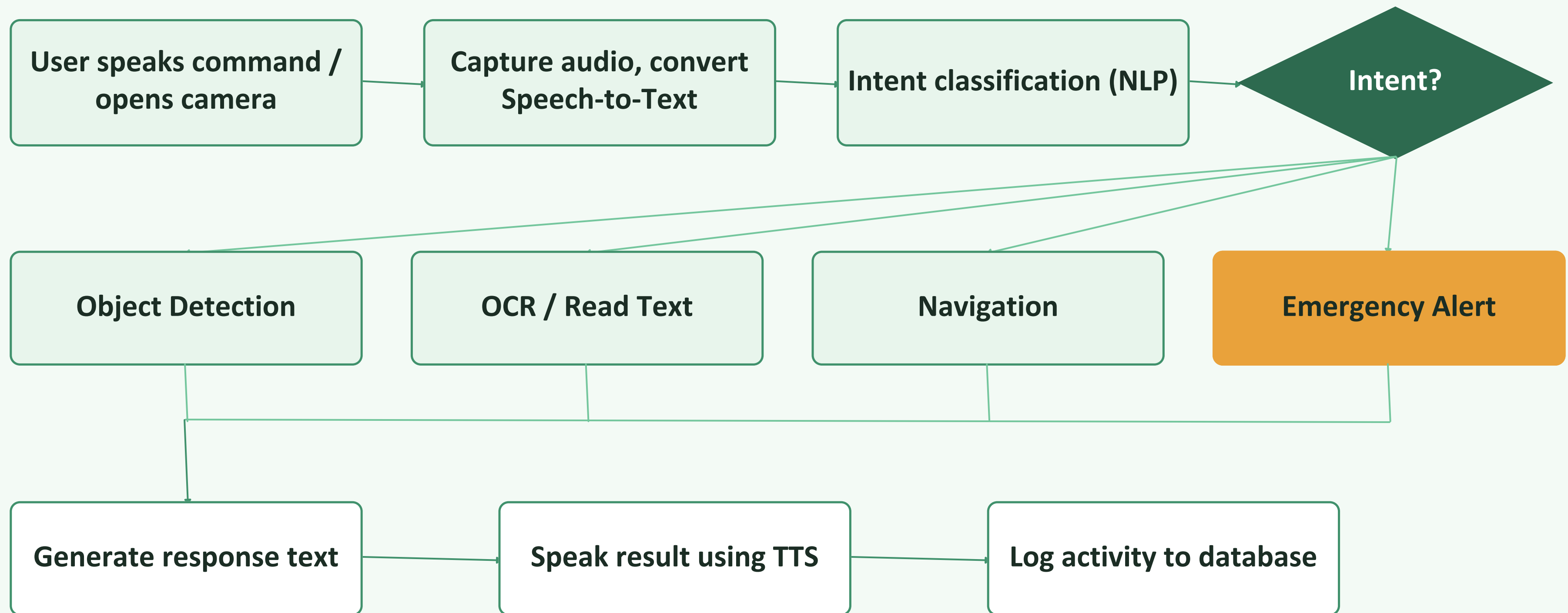
Use Case Diagram



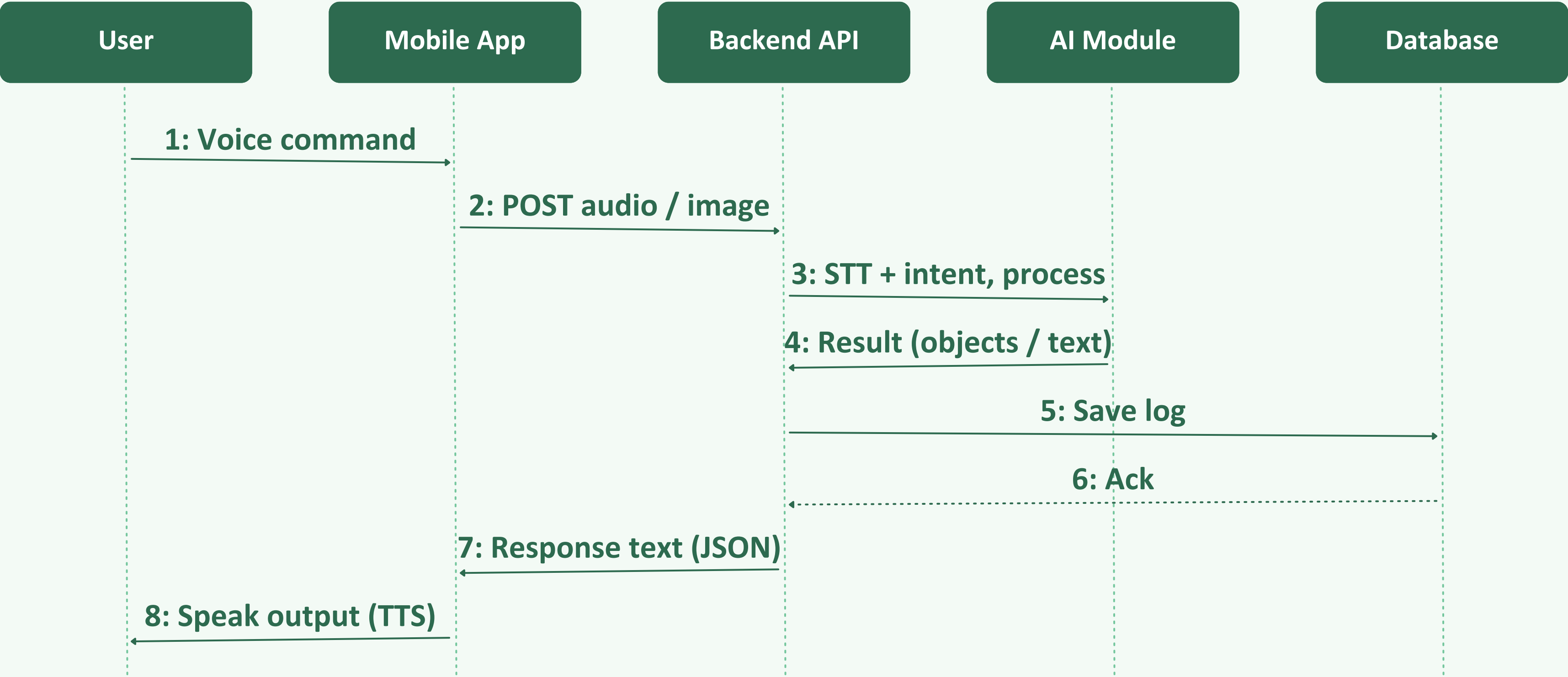
Class Diagram



Activity Diagram



Sequence Diagram



Screen Layouts



1. Splash

Logo

Voice welcome: "Hello, I am VisionMate"

2. Home

Big mic button

Detect | Read | Navigate,
SOS at bottom

3. Detect Objects

Live camera view

Speaks object + distance, tap
to repeat

4. Text Reader

Capture page / label

Reads aloud, pause / speed

5. Navigation

Say destination

Turn-by-turn voice, saved
places

6. Emergency SOS

Huge red button

Sends location to guardian

7. Settings

Language (Tamil / English)

Voice speed, pitch, guardian
contacts

8. History

Past commands

Read-aloud list

Module Allocation



- **M1 Voice:** STT, TTS - [Member 1] []%
- **M2 Vision:** YOLOv8, OCR - [Member 2] []%
- **M3 NLP + Backend:** [Member 3] []%
- **M4 App UI + Safety:** [Member 4] []%
- **Phase I status:** [update with actual progress]
- **Review I actions:** [write recommendations and updates]

GitHub Repository



- **Repo:** visionmate-ai
- **Branches:** voice, vision, backend, ui
- **Folders:** app, backend, ai, database, docs, tests
- Merge through pull requests
- Visible commits for every member

Research Paper



- **Title:** VisionMate AI: Voice-Driven Multimodal Assistant for the Visually Impaired
- **Gap:** no unified voice-first system; limited Tamil support
- **Method:** STT, intent, YOLOv8, OCR, SOS; measure mAP, WER, latency, SUS

Expected Results



> 0.80

mAP for object
detection

> 90%

OCR accuracy on
printed text

2-3 s

end-to-end
response

Publication Plan



- **Target:** IEEE / Springer conference + backup journal
- **Journals:** Assistive Technology, IEEE Access
- **Format:** IEEE two-column, 6-8 pages
- **Status:** [Drafting / Submitted]
- **Timeline:** draft Nov 2026, submit Feb-Mar 2027

Patent Disclosure Draft



- **Title:** Voice-controlled multimodal assistive system
- One command routed to the right AI module
- English + Tamil, including code-mixed commands
- Hazard-aware spoken scene summary
- One-command SOS with live location

Novelty Assessment



Feature	Existing apps	VisionMate AI
Voice-first control	Partial	Full hands-free
Object detection	Some	Yes, spoken priority
Text reading (OCR)	Some	Yes
Navigation by voice	Separate apps	Integrated
Emergency alert + location	Rare	One command
Tamil support	Limited	English + Tamil
Extra hardware	No	No (phone only)

Phase II Implementation Plan



Month 1
Jan 2027

Datasets, STT / TTS, intent, API

Month 2
Feb 2027

YOLOv8 and OCR, fine-tuning, Flutter screens

Month 3
Mar 2027

Navigation, emergency alert, full integration

Month 4
Apr 2027

Testing, user validation, deployment, paper

Training, Proposals, Hackathon



- **Industrial training:** Pending - [organization, dates]
- **Proposal:** [Name] - [Submitted / Planned]
- **Hackathon:** [Name] - [Prototype progress]